

WORK SHEET - 03

VECTOR & 3D

[HINTS & SOLUTIONS]

Q.1 18

Sol. $\vec{V}_1 \cdot \vec{V}_2 = 0 \Rightarrow a + b + c = 0$

$$\frac{a}{-2} \quad \frac{b}{1} \quad \frac{c}{1} \Rightarrow \frac{3!}{2!} = 3$$

$$\frac{-2}{-2} \quad \frac{0}{0} \quad \frac{2}{2} \Rightarrow \frac{3!}{1!} = 6$$

$$\frac{-1}{-1} \quad \frac{0}{0} \quad \frac{1}{1} \Rightarrow \frac{3!}{1!} = 6$$

$$\frac{-1}{-1} \quad \frac{-1}{-1} \quad \frac{2}{2} \Rightarrow \frac{3!}{2!} = 3$$

Total number of vectors = 18

Q.2 $\left| \frac{3k^2}{\sqrt{1+k^2+k^4}} \right|$

Sol. Equation of plane is given by $\frac{x}{3} + \frac{y}{3k} + \frac{z}{3k^2} = 1$ or $k^2x + ky + z = 3k^2$.

$\Rightarrow$ Distance of plane from origin equal to $\left| \frac{3k^2}{\sqrt{1+k^2+k^4}} \right|$

Q.3 -10

Sol. $\therefore (\hat{a} \times (\hat{b} \times \hat{c})) \cdot (\hat{a} \times \hat{c}) = 5$

$$\Rightarrow ((\hat{a} \cdot \hat{c})\hat{b} - (\hat{a} \cdot \hat{b})\hat{c}) \cdot (\hat{a} \times \hat{c}) = 5$$

$$\Rightarrow (\hat{a} \cdot \hat{c})(\hat{b} \cdot (\hat{a} \times \hat{c})) - 0 = 5 \Rightarrow \frac{1}{2}[\hat{b} \hat{a} \hat{c}] = 5$$

$$\Rightarrow [\hat{a} \hat{b} \hat{c}] = -10$$

Q.4 least value of $\vec{a} \cdot \vec{b} + \frac{4}{|\vec{b}|^2 + 2}$ is $2\sqrt{2} - 1$.

Sol. $2\vec{a} \cdot \vec{b} = -4\vec{a} \cdot \vec{b} + 3\vec{b}^2$

$$2\vec{a} \cdot \vec{b} = \vec{b}^2$$

$$\Rightarrow \vec{a} \cdot \vec{b} + \frac{4}{|\vec{b}|^2 + 2} = (1 + \vec{a} \cdot \vec{b}) + \frac{4}{2(\vec{a} \cdot \vec{b} + 1)} - 1 \geq 2\sqrt{2} - 1$$

Q.5 3

Sol. Given line 'L' makes 45° with the given plane

$$\text{Hence, } PQ = \frac{|\sqrt{2} + 7 - \sqrt{2} - 1|}{\sqrt{2+1+1}} = \frac{6}{2} = 3$$

Q.6 $\frac{2}{3}$

Sol. Let $\vec{OA} = \hat{a}$, $\vec{OB} = \hat{b}$, $\vec{OC} = \hat{c}$ and $\vec{OP} = \hat{d}$

Here $\hat{a} \cdot \hat{b} = \hat{b} \cdot \hat{c} = \hat{c} \cdot \hat{a} = \frac{1}{2}$

and $\hat{d} \cdot \hat{a} = \hat{d} \cdot \hat{b} = \hat{d} \cdot \hat{c} = \cos \alpha$

Let $\hat{d} = x\hat{a} + y\hat{b} + z\hat{c}$ (i)

$$\hat{d} \cdot \hat{a} = x + \frac{y}{2} + \frac{z}{2} = \cos \alpha$$

$$\hat{d} \cdot \hat{b} = \frac{x}{2} + y + \frac{z}{2} = \cos \alpha$$

$$\hat{d} \cdot \hat{c} = \frac{x}{2} + \frac{y}{2} + z = \cos \alpha$$

Adding we get $2(x + y + z) = 3 \cos \alpha$

Taking dot product with $\hat{d}$ in (i) we get

$$1 = x \cos \alpha + y \cos \alpha + z \cos \alpha \Rightarrow x + y + z = \frac{1}{\cos \alpha}$$

$$\Rightarrow \frac{3 \cos \alpha}{2} = \frac{1}{\cos \alpha} \Rightarrow \cos^2 \alpha = \frac{2}{3}$$

Q.7 $\frac{1}{\sqrt{2}}$

Sol. Let three orthonormal vectors are $\hat{i}$, $\hat{j}$ and $\hat{k}$ then

$$\vec{\alpha} = \frac{\hat{i} + \hat{j}}{\sqrt{2}}, \vec{\beta} = \frac{\hat{j} + \hat{k}}{\sqrt{2}}, \vec{\gamma} = \frac{\hat{i} + \hat{k}}{\sqrt{2}}$$

$$\Rightarrow \text{volume} \left[\frac{\hat{i} + \hat{j}}{\sqrt{2}}, \frac{\hat{j} + \hat{k}}{\sqrt{2}}, \frac{\hat{i} + \hat{k}}{\sqrt{2}} \right] = \frac{1}{2\sqrt{2}} \times 2[\hat{i} \hat{j} \hat{k}] = \frac{1}{\sqrt{2}}$$

Q.8 (1, 1)

Sol. We must have $\begin{vmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 1 & x & y \end{vmatrix} = 0$

$$\Rightarrow x - 3y + 2 = 0 \quad \text{.....(1)}$$

Also, $x^2 + y^2 + 1 = 3$

$$\Rightarrow x^2 + y^2 = 2 \quad \text{.....(2)}$$

$$\therefore (1) \cap (2)$$

$$\Rightarrow (1, 1)$$

Q.9 $6\sqrt{13}$ (D) $10\sqrt{13}$

Sol. $\gamma = 12 \sin \theta + 18 \cos \theta$ where $\theta = \vec{\alpha} \wedge \vec{\beta}$

$$\therefore \gamma \leq \sqrt{(12)^2 + (18)^2} = \sqrt{468} = \sqrt{36 \times 13} = 6\sqrt{13}$$

Q.10 1

Sol. $|\vec{p} \times (\vec{q} - \vec{r})|^2 = |\vec{p}|^2 |\vec{q} - \vec{r}|^2 - (\vec{p} \cdot (\vec{q} - \vec{r}))^2 = |\vec{p}|^2 |\vec{q} - \vec{r}|^2 = (1)^2 \left(1 + 1 - 2 \cos \frac{\pi}{3}\right) = 1$

Q.11 $\frac{7}{3\sqrt{6}}$

Q.12 $\frac{3}{\sqrt{6}}$

Q.13 $2x - y + 2z = 0$

Sol. $\vec{r}_1 \cdot \vec{r}_2 = 0$

$$2a^3 + 2ab + 6a^2 = 0$$

$$a^2 + b + 3a = 0$$

$$b = -a^2 - 3a = \frac{9}{4} - \left(a + \frac{3}{2}\right)^2 \Rightarrow b \leq \frac{9}{4}$$

$\therefore$ largest integral value of b is 2

Now, for b = 2, $a^2 + 3a + 2 = 0 \Rightarrow a = -1, -2$

For $a = -1, b = 2$

$$\vec{r}_1 = \hat{i} - 4\hat{j} - 6\hat{k}$$

$$\vec{r}_2 = -2\hat{i} + \hat{j} - \hat{k}$$

$$\vec{r}_3 = -\hat{i} - 2\hat{j} - \hat{k}$$

For $a = -2, b = 2$

$$\vec{r}_1 = 4\hat{i} - 4\hat{j} - 6\hat{k}$$

$$\vec{r}_2 = -4\hat{i} + 2\hat{j} - 4\hat{k}$$

$$\vec{r}_3 = \hat{i} - 2\hat{j} - 2\hat{k}$$

Plane P_1

$$-2x + y - z + \lambda = 0$$

(1, -2, -1),

$$-2 - 2 + 1 + \lambda = 0$$

$$\Rightarrow \lambda = 3$$

$$\therefore P_1 : -2x + y - z + 3 = 0$$

Plane P_2

$$-4x + 2y - 4z + \lambda' = 0$$

(1, -2, -2),

$$-4 - 4 + 8 + \lambda' = 0$$

$$\Rightarrow \lambda' = 0$$

$$\therefore P_2 : 2x - y + 2z = 0$$

(i) $\cos \theta = \left| \frac{(-2)(-4) + 1(2) + (-1)(-4)}{\sqrt{6} \sqrt{14 + 4 + 16}} \right| = \frac{14}{6\sqrt{6}} = \frac{7}{3\sqrt{6}}$

(ii) $d_1 = \frac{3}{\sqrt{6}}, \quad d_2 = 0$

$$\therefore d_1 + d_2 = \frac{3}{\sqrt{6}}$$



- (iii) $p_1 + \lambda p_2 = 0$
 $-2x + y - 3 + 3 + \lambda(2x - y + 2z) = 0$
 which passes through $(0, 0, 0)$
 $3 + 0 = 0$ (absurd result)
 It means p_2 is the required plane.

Q.14 Statement-1 is false, statement-2 is true.

Sol. Statement-1 : Lines L_1, L_2 and L_3 are parallel
 Plane containing lines L_1 & L_2 is

$$\begin{vmatrix} x-1 & y-3 & z-2 \\ 1 & 2 & -3 \\ 1/2 & 3 & 1 \end{vmatrix} = 0$$

L_3 passing through the point $(2, 5, 5)$

$$\therefore \begin{vmatrix} 1 & 2 & 3 \\ 1 & 2 & -3 \\ 1/2 & 3 & 1 \end{vmatrix} \neq 0$$

Hence, lines L_1, L_2 and L_3 are not coplanar.

Q.15 Statement-1 is false, statement-2 is true.

Sol. Since $\vec{a} \cdot (\vec{b} \times \vec{c}) = 0 = [\vec{a} \vec{b} \vec{c}]$

$\Rightarrow \vec{OA}, \vec{OB}$ and $\vec{OC}$ are coplanar.

So tetrahedron is not formed.

Q.16 Statement-1 is true, Statement-2 is false.

Sol. S-1 is true. $(\vec{r} - \vec{a}) \times \vec{b} = \vec{0}$ is the equation of the straight line passing through the point whose position vector is $\vec{a}$ and the vector $\vec{b}$ lies along the line. Again since vector $\vec{b}$ is perpendicular to the plane

$$(\vec{r} - \vec{a}) \cdot \vec{b} = 0$$

$\therefore$ the line is perpendicular to the given plane.

$\therefore$ Statement-1 is true.

Statement-2: $\vec{r} \cdot \vec{b} = \vec{a} \cdot \vec{b} \Rightarrow (\vec{r} - \vec{a}) \cdot \vec{b} = 0$ represent a plane.

If point $\vec{b}$ lies on the plane then $\vec{b} \cdot \vec{b} = \vec{a} \cdot \vec{b} \Rightarrow \vec{a} \cdot \vec{b} = 1$ which is not true.

$\therefore$ Statement-2 is false.

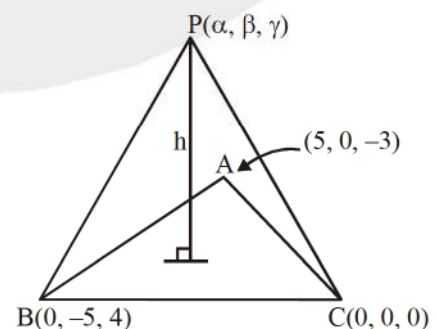
Q.1 (A) S; (B) R; (C) P; (D) P

Sol. $\vec{BC} \times \vec{BA} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 5 & -4 \\ 5 & 5 & -7 \end{vmatrix}$

$$= \hat{i}(-15) - \hat{j}(20) + \hat{k}(-25)$$

$$= -15\hat{i} - 20\hat{j} - 25\hat{k}$$

$$\text{Area } (\Delta ABC) = \frac{1}{2} \sqrt{(15)^2 + (20)^2 + (25)^2}$$



$$= \frac{1}{2} \sqrt{225 + 400 + 625} = \frac{1}{2} \cdot 25\sqrt{2} = \frac{25}{\sqrt{2}}$$

Volume of tetrahedron $V = \frac{1}{3} \times \frac{25}{\sqrt{2}} \times h = 25 \Rightarrow h = 3\sqrt{2}$

Locus of P is the plane parallel to the given plane at a distance of $3\sqrt{2}$ units

$$3x + 4y + 5z + d = 0$$

$$\therefore \left| \frac{d}{\sqrt{9+16+25}} \right| = 3\sqrt{2} \Rightarrow d = \pm 30$$

$$\therefore 3x + 4y + 5z \pm 30 = 0$$

(P) $k = 30$

(Q) Area of parallelogram ABCD = $\left| \vec{BC} \times \vec{BA} \right| = 25\sqrt{2}$

(R) Least value of PG = $h = 3\sqrt{2}$

(S) Least value of PH = $h = 3\sqrt{2}$

Q.18 [2]

Sol. Let M be foot of perpendicular from Q on plane : drs of QM $\Rightarrow 1, 2, 1$
equation of QM is

$$\frac{x-3}{1} = \frac{y-5}{2} = \frac{z-2}{1} = \lambda$$

A point M on line can be taken as $(\lambda + 3, 2\lambda + 5, \lambda + 2)$

$\therefore$ M lies on the plane.

$$\therefore (\lambda + 3) + 2(2\lambda + 5) + (\lambda + 2) - 3 = 0 \Rightarrow 6\lambda = -12 \Rightarrow \lambda = -2$$

$$\therefore M = (1, 1, 0)$$

Let image of Q is plane be $Q'(x_1, y_1, z_1)$

$$\therefore \frac{x_1+3}{2} = 1, \frac{y_1+5}{2} = 1, \frac{z_1+2}{2} = 0. \therefore Q' = (-1, -3, -2)$$

$$\therefore PR + QR = PR + Q'R$$

$PR + Q'R$ is min.

$\therefore$ R must be point of intersection of line with the plane.

$$\therefore \text{equation of } PQ' \Rightarrow \frac{x-1}{2} = \frac{y-3}{6} = \frac{z-2}{4} \Rightarrow \frac{x-1}{1} = \frac{y-3}{3} = \frac{z-2}{2} = \mu$$

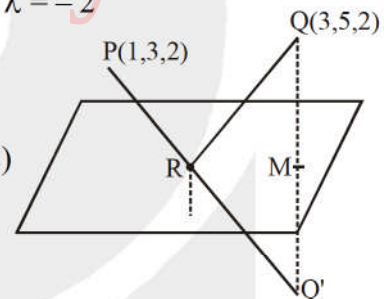
Any point R on this line can be taken as $(\mu + 1, 3\mu + 3, 2\mu + 2)$

$\therefore$ Point R on the plane. $\therefore$ It will satisfy its equation.

$$\therefore (\mu + 1) + 2(3\mu + 3) + (2\mu + 2) - 3 = 0 \Rightarrow 9\mu = -6 \Rightarrow \mu = -\frac{2}{3}$$

$$\therefore R \left(\frac{1}{3}, 1, \frac{2}{3} \right) = (a, b, c)$$

$$\therefore a + b + c = 2.$$



Q.19 [1]

Sol. $\vec{v}_1 = 3(\vec{a} \times \vec{b}), \vec{v}_2 = 2(\vec{b} - (\vec{a} \cdot \vec{b})\vec{a})$

$$\vec{v}_1 \cdot \vec{v}_2 = 0 \Rightarrow \vec{v}_1 \perp \vec{v}_2$$

$$|\vec{v}_1| = 3|\vec{a}| |\vec{b}| \sin \theta = 3|\vec{b}| \sin \theta$$

$$|\vec{v}_2| = 2\sqrt{(\vec{b} - (\vec{a} \cdot \vec{b})\vec{a})^2} = 2\sqrt{|\vec{b}|^2 - 2(\vec{a} \cdot \vec{b})^2 + (\vec{a} \cdot \vec{b})^2 |\vec{a}|^2}$$

$$= 2\sqrt{|\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2}$$

$$= 2\sqrt{|\vec{b}|^2 - a^2 b^2 \cos^2 \theta} = 2|b| \sin \theta$$

$$\text{Area} = \frac{1}{2} \cdot 3|\vec{b}| \sin \theta \cdot 2|\vec{b}| \sin \theta = \frac{3}{4}(|\vec{b}|^4 + 4)$$

$$|b|^2 \sin^2 \theta = \frac{1}{4}(|b|^4 + 4)$$

$$\underbrace{4\sin^2 \theta}_{\leq 4} = \underbrace{|\vec{b}|^2 + \frac{4}{|\vec{b}|^2}}_{\geq 4}$$

$$|\vec{b}| = \sqrt{2}$$

$$\therefore [|\vec{b}|] = 1$$